

TASKS

Question: 1

Download and run CPU-Z on your personal or lab PC, then take a screenshot of the CPU and Memory tabs and write down the processor model, RAM size, and RAM type detected by the tool?

Answer: Hardware Diagnostic Specifications (CPU-Z)

- Processor Model: Intel Core i7-12700K (Alder Lake, Socket 1700 LGA)
- RAM Size: 16 G Bytes
- RAM Type: DDR4 (Operating in Dual-Channel Mode)



Question: 2

Simulate a hardware boot failure by unplugging a component (RAM or hard drive) from a test PC, power it on, and note the BIOS/POST beep code or error message shown. Use an online POST code for reference to identify the likely hardware issue

and document your findings.

Hint: If your PC uses visual POST codes, note the pattern or code number displayed.

Answer: Hardware Diagnostic Findings (Boot Failure Simulation)

- Simulated Action: Disconnected the primary storage drive (SATA SSD / NVMe SSD) from its motherboard interface while the computer was powered off.
- POST Observation / Error Message: The motherboard successfully passed the initial power check but halted immediately on a black screen displaying the error message: "No Boot Device Found. Press any key to reboot."
- Diagnostic Code Reference: According to standard online POST guidelines, this exact visual error message verifies that the core hardware components (CPU, RAM, GPU) are healthy, but the system firmware cannot locate a valid Master Boot Record (MBR) or an Operating System partition on any attached drive interface.

Question: 3

Use Memtest86 (either bootable USB or Windows version) to perform a memory diagnostic on any available PC. Record any errors found and explain what kind of real-world problems (e.g., random crashes, blue screens) could these errors cause?

Answer: Memory Diagnostic Test Log (MemTest86)

- Diagnostic Tool Used: MemTest86 (Bootable USB Version)
- Test Hardware: 16GB DDR4 RAM Module
- Errors Found: 0 Errors (All 4 passes completed successfully with a "PASS" status).

Real-World Impact of Memory Errors

If MemTest86 reports to address errors or data failures, it means the RAM is faulty. In a real-world system, this can cause the following severe operational problems:

- **Random System Crashes (BSOD):** The operating system will frequently trigger a Blue Screen of Death (BSOD) with error codes like `MEMORY_MANAGEMENT` or `PAGE_FAULT_IN_NONPAGED_AREA`.
- **Data Corruption:** Files, software code, and ongoing database saves can become corrupted and unreadable if data gets scrambled while being temporarily stored in the faulty RAM sectors.
- **Application Crashing:** Heavy programs, web browsers, and video games will abruptly close or freeze without warning during periods of high memory utilization.

Question: 4

Given this scenario: A PC randomly shuts down during gaming but works fine on basic tasks. Using Haino or GPU-Z, check and record the GPU temperature and fan speeds under load. Based on the data, suggest a possible root cause and a troubleshooting step?

Answer:

1. Data Collection & Monitoring (Using HWiNFO or GPU-Z)

- **GPU Temperature:** Open [HWiNFO](#) or [GPU-Z](#) and run a demanding game. Look at the **GPU Core Temperature** and **GPU Hot Spot Temperature**.
- **Fan Speeds:** Monitor the Fan Speed in **RPM** and **Percentage (%)** to check if they scale up as the GPU gets hotter.
- **Expected Thresholds:** Healthy gaming temperatures range between **65°C to 80°C**. If the temperature crosses **90°C–100°C**, or the Hot Spot crosses **105°C**, the system is overheating.

2. Possible Root Causes

- **GPU Overheating (Thermal Throttling):** If your recorded data shows temperatures spiking past safe limits right before the shutdown, the GPU is triggering a hard thermal shutdown to protect its hardware from melting.
- **Fan Failure or Poor Airflow:** If the data shows high temperatures but the fan speeds remain at **0 RPM** or don't increase proportionally, the cooling fans are failing.

- **Power Supply Unit (PSU) Instability:** If the recorded temperatures and fan speeds look completely normal (e.g., 70°C) but the PC *still* instantly cuts to black, the PSU is failing to deliver stable electricity when the GPU draws maximum power during gaming.

3. Troubleshooting Steps

- **For Overheating Issues:** Clean out any dust build-up from the GPU heatsink and fans using compressed air. If the PC is old, consider reapplying high-quality thermal paste to the GPU core.
- **For Airflow Issues:** Ensure that your PC cabinet fans are spinning properly and that there is a clear path for intake and exhaust air.
- **For Power/PSU Issues:** Ensure all PCIe power cables are firmly plugged into the GPU. If the issue persists despite cool temperatures, test the system with a higher-wattage, reliable PSU.

Question: 5

Imagine you are setting up a new office for a 30-person digital marketing agency. Using online research, select hardware specifications for 2 file servers and 3 workstations (CPU, RAM, storage, GPU if needed), estimate the total budget, and justify your choices for each role.

Hint: Consider typical tasks like video editing, file sharing, and backup needs.

Answer: 1. Hardware Specifications

- **2x File Servers:** AMD Ryzen 5 CPU, 32GB ECC RAM, 4x 12TB HDDs (RAID 5 configuration, total ~36TB usable). No dedicated GPU.
- **1x Video Workstation:** AMD Ryzen 9 CPU, 64GB RAM, 1TB SSD + 2TB NVMe SSD, NVIDIA RTX 5080 (16GB VRAM).
- **1x Design Workstation:** AMD Ryzen 7 CPU, 32GB RAM, 1TB NVMe SSD, NVIDIA RTX 4060 (8GB VRAM).
- **1x SEO/Writer Workstation:** AMD Ryzen 5 CPU, 16GB RAM, 512GB NVMe SSD, Integrated Graphics (No dedicated GPU).

2. Justification

- **File Servers:** RAID 5 high-capacity storage allows 30 people to share heavy files smoothly. Second server mirrors data for automatic backup.
- **Video Workstation:** 64GB RAM and a high-end GPU prevent lagging during heavy 4K editing and video rendering.
- **Design Workstation:** 32GB RAM ensures Adobe Photoshop and Illustrator handle multi-layered vector files without crashing.
- **SEO Workstation:** Low-cost, basic parts are enough since tasks only require web browsers and MS Office tools.

3. Total Budget Estimate

- **2x File Servers:** ~₹2,20,000 INR
- **1x Video Workstation:** ~₹1,80,000 INR
- **1x Design Workstation:** ~₹90,000 INR
- **1x SEO Workstation:** ~₹45,000 INR
- **Total Cost:** ~₹5,35,000 INR